

SUMMER TRAINING REPORT (ICT-393)

Submitted in partial fulfillment of the requirements for the award of the degree of

**BACHELOR OF TECHNOLOGY
IN
COMPUTER SCIENCE AND ENGINEERING**



**UNIVERSITY SCHOOL OF INFORMATION
AND
COMMUNICATION TECHNOLOGY**

**GURU GOBIND SINGH INDRAPRASTHA
UNIVERSITY, NEW DELHI-110078**

(2023-2027)

SUBMITTED BY:

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BRANCH: B.TECH CSE 5th Sem

DECLARATION

I, Manish Rouniyar (40316403223), a student of Bachelor of Technology in Computer Science and Engineering, hereby declare that the internship report titled “Frontend Developer Internship at Eknal Technologies Pvt. Ltd.” is an original work completed by me.

This report is a reflection of the tasks and projects I undertook during my summer internship from 17th April 2025 to 18th July 2025 as a Frontend Developer Intern. The information presented here is true to the best of my knowledge and has not been copied or submitted for any other purpose.

I have adhered to academic and ethical standards throughout the preparation of this report. Any assistance, guidance, and data obtained from external sources have been duly acknowledged.

LOCATION: Andhra Pradesh, India

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BRANCH: B.TECH CSE 5th Sem

Acknowledgment

I would like to express my heartfelt gratitude to Eknaal Technologies Pvt. Ltd. and Mr. Ramesh Lanke (mentor & Co-Founder) for giving me the opportunity to work under their guidance and helping me gain an immensely enriching professional experience. I sincerely thank my mentor and teammates for giving me valuable inputs and ideas right from the selection of the topic for the project till its successful completion.

I also extend my heartfelt thanks to my professors and internship coordinators at USICT for their constant encouragement and for equipping me with the foundational knowledge and skills needed to succeed in this internship.

The successful completion of our project would not have been possible without the dedicated support from all my mentors, family, and friends.

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INTERNSHIP CERTIFICATE



Traycer AI, Inc.

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1-215-919-0137

Date: 11/01/2024

Certificate of Completion

This is to certify that **Anurag Sharma** has successfully completed a two-month internship program at **Traycer AI, Inc.** from **July 8th, 2024** to **September 8th, 2024**.

During this period, he actively contributed to development and enhancement of code analysis tools which are powered by Artificial Intelligence. He demonstrated professionalism, diligence, and commitment in all assigned tasks. His performance was commendable, and he consistently showed a willingness to learn and adapt to new challenges.

We wish Anurag Sharma all the best in his future endeavors.

Tanveer Gill

CEO

Traycer AI, Inc.

1. Introduction and Company Profile

1.1 Overview of the Internship

During my summer internship at Traycer AI, Inc., I focused on front-end development for the Memories feature, utilizing Retrieval-Augmented Generation (RAG) technology to enhance the user experience in Traycer's AI-powered code analysis tool. Working closely with the engineering team, I developed React components to seamlessly display memory-based insights, creating an intuitive and responsive interface. My work involved designing interactive UI elements that enabled developers to easily access and manage contextual suggestions within their coding environment. This experience strengthened my skills in crafting user-friendly interfaces for AI-driven tools, focusing on delivering a smooth, productive experience for end-users.

1.2 About Traycer AI, Inc.

Traycer AI, Inc. is an innovative software company dedicated to enhancing code quality and developer productivity through intelligent automation. Its flagship product is an advanced code analysis tool that integrates directly into Visual Studio Code, providing real-time, context-aware feedback to developers. This integration allows for seamless identification and resolution of potential issues, streamlining the development process.

1.3 Key Products and Services

Traycer's primary offering is its AI-powered code analysis tool, which supports multiple programming languages, including TypeScript, JavaScript, Python, and Golang. Key features include automatic tracking of code changes, actionable feedback, single-click fixes, and context-aware conversations with AI for instant clarifications and code improvements. These features are designed to keep developers in the flow, analyzing code changes in the background and delivering clear and actionable insights without interrupting their workflow.

1.4 Company Details

Website: <https://traycer.ai>

Headquarters: 376 La Casa Via, Walnut Creek, California 94598, USA

2. Project Focus and Responsibilities

2.1 Feature Overview: Memories with RAG Integration

The Memories project involved designing and refining a front-end interface within Traycer AI, using Retrieval-Augmented Generation (RAG) technology. This feature was intended to enhance developer productivity by integrating contextual insights based on prior interactions, allowing developers to access relevant suggestions directly within the UI for more efficient coding.

2.2 UI Objectives

The primary objectives for the UI were to create a React-based, intuitive interface for memory retrieval, facilitating easy access to contextual information. The project emphasized integrating RAG seamlessly to ensure the feature delivered relevant insights while maintaining a responsive and efficient user experience.

2.3 Scope and Milestones

The project's scope encompassed developing front-end components for memory display, designing a user-friendly layout, and optimizing the UI for speed and reliability. Key milestones included building the initial UI layout, integrating RAG-based retrieval, and conducting usability testing to achieve a smooth, cohesive interface within the internship timeline.

2.4 Contributions to Front-End Development

I contributed by implementing React components that display memory retrieval results within the Traycer UI, focusing on enhancing the flow of contextual insights. I optimized the front-end to interact efficiently with the RAG system, collaborated with the team on troubleshooting, and refined the overall user experience. My responsibilities also included validation testing and documenting UI elements to support future updates.

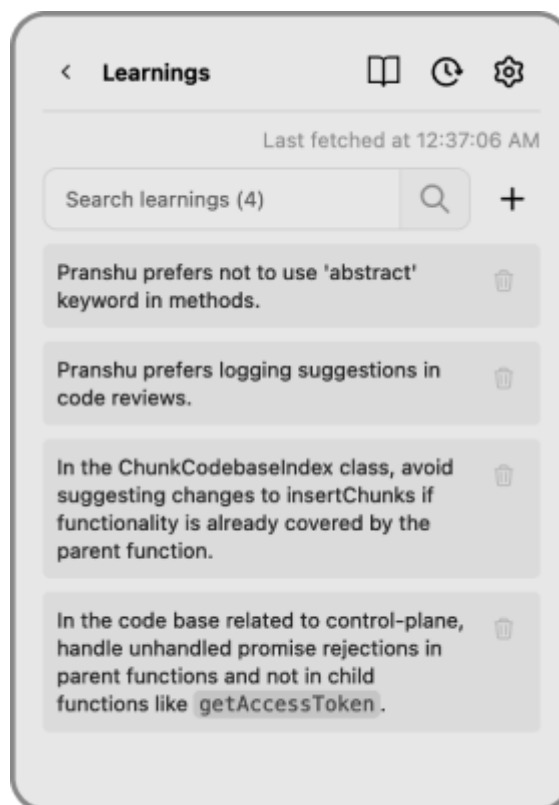
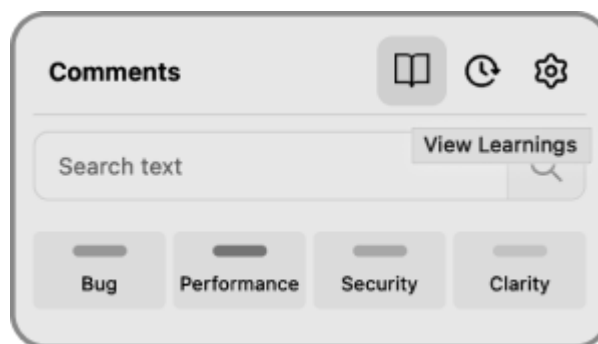
2.5 Tools and Frameworks

The project utilized React for component development, with RAG frameworks for retrieving memory data, GitHub for version control, and debugging tools within the IDE. This toolkit enabled effective front-end development, allowing for testing and optimization of UI performance throughout the project's phases.

3. Technical Approach

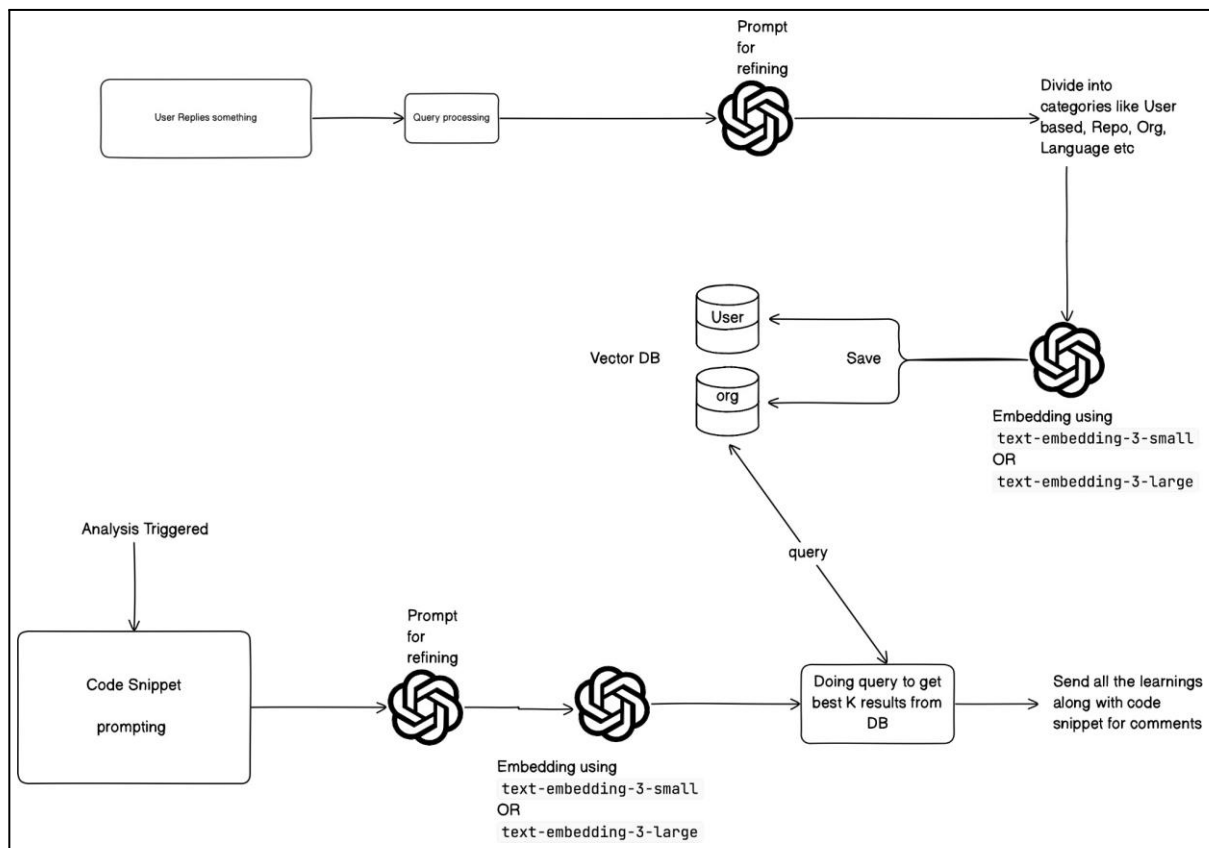
3.1 Overview of RAG (Retrieval-Augmented Generation) System

The Memories feature leverages a Retrieval-Augmented Generation (RAG) system to provide contextually relevant suggestions within the front-end interface. RAG combines retrieval and generation techniques, pulling relevant information from prior interactions or documentation and presenting it in a way that enhances the current coding context. This setup enables the front-end to display meaningful, context-aware insights directly within the IDE, helping developers quickly access relevant suggestions and improve their workflow without switching contexts. The RAG-driven UI elements make the user experience seamless, displaying suggestions in real-time and in alignment with the user's current development focus.



3.2 Development Process

The development process involved key stages focused on front-end integration and performance optimization: initial UI design, embedding the retrieval mechanisms, testing for response accuracy, and optimizing for speed. In the interface, Pinecone was integrated as the vector database to efficiently store and retrieve embeddings, enabling fast access to relevant memories. OpenAI's large embedding model generated high-context embeddings that seamlessly aligned with the front-end display, ensuring suggestions were highly relevant. Additionally, Supabase served as the data storage layer, providing a reliable backend for managing retrieval operations and supporting a smooth user experience within the IDE.



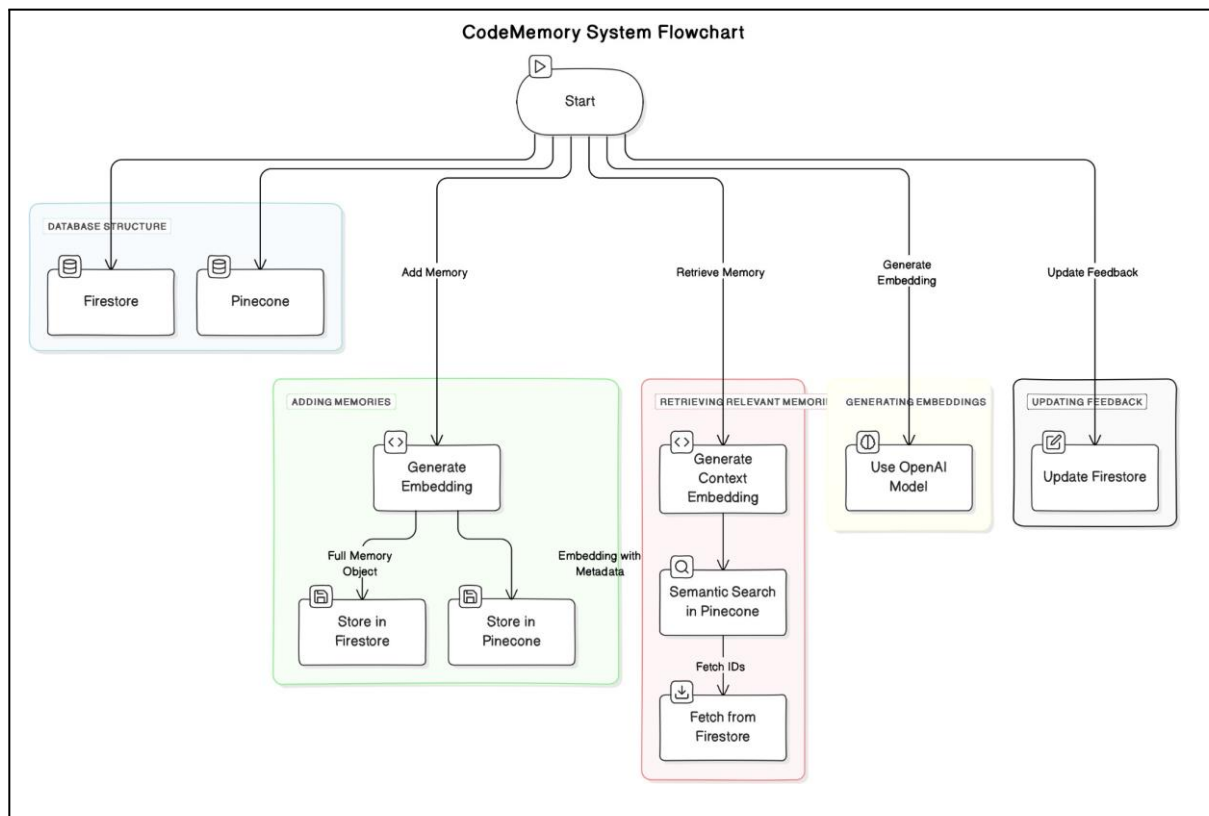
3.3 Challenges and Solutions

Ensuring a responsive UI with real-time data retrieval was a primary challenge, requiring efficient loading indicators and asynchronous data handling to avoid delays. Managing large embeddings from OpenAI while maintaining quick access in Pinecone also tested front-end performance. Displaying context-relevant suggestions without UI lag involved careful optimization of component rendering. These adjustments were crucial to delivering a seamless and intuitive user experience within the IDE.

3.4 Implementation Details

The RAG system was configured to retrieve the most relevant matches, with Pinecone storing and searching vectorized representations of past interactions to support quick access within the UI. Supabase served as the primary data store, managing and tracking interactions to maintain memory consistency. Leveraging OpenAI's large embedding model, the system generated embeddings that captured the semantic context of past interactions, ensuring that the suggestions displayed in the UI were both meaningful and specific to the developer's current coding context. This configuration provided developers with targeted, context-aware insights seamlessly integrated into their IDE.

3.5 Flowchart



4. Achievements and Learning Outcomes

4.1 Front-End Accomplishments

During my internship, I played a key role in developing the front-end for the Memories feature, focusing on seamless integration of RAG-based suggestions within the UI. This involved configuring the retrieval system for efficient data display and optimizing React components to ensure relevant suggestions appeared quickly without lag. I also fine-tuned how embeddings were presented to enhance the accuracy and quality of insights, improving the overall user experience and boosting developer productivity within Traycer AI.

4.2 Technical Skills Gained

This experience allowed me to deepen my front-end skills, especially in React and state management for real-time data retrieval. I gained practical experience with Pinecone for efficient data handling, OpenAI's embeddings for delivering contextual insights, and Supabase for managing interactions. Additionally, I learned best practices for implementing AI-driven features in user interfaces within IDEs, developing collaborative problem-solving skills by working closely with other developers to troubleshoot and refine the user experience.

4.3 Professional and Personal Growth

Professionally, I gained confidence in tackling complex UI challenges and learned to approach project tasks with a proactive mindset. This experience also strengthened my collaboration skills, helping me communicate technical details effectively and integrate feedback for continuous improvement. Personally, this internship reinforced my enthusiasm for AI-driven front-end development and my commitment to advancing my skills in creating intuitive, high-performance user interfaces.

5. Conclusion

5.1 Overview of Experience

My summer internship at Traycer AI, Inc. was a deeply rewarding opportunity that enabled me to work on the Memories feature, an impactful addition designed to improve developer productivity. Engaging with Retrieval-Augmented Generation (RAG) technology, I was able to put my skills into action, tackling technical challenges, and gaining hands-on experience with AI-driven front-end development. This project underscored the importance of creating a seamless user experience, blending advanced functionality with practical usability.

5.2 Insights and Takeaways

Throughout this internship, I gained valuable insights into applying AI technologies within user interfaces, emphasizing the balance between technical efficiency and a user-friendly experience. The collaborative environment at Traycer AI fostered both creativity and personal growth, with team support that encouraged continuous learning and innovation. This experience enhanced my understanding of the role AI can play in streamlining coding workflows, as well as the importance of clear, efficient UI design.

5.3 Future Potential for the Memories Feature

Looking forward, there are numerous possibilities for expanding the Memories feature to support an even broader range of developer needs. Potential improvements could include advanced filtering options for retrieving context-specific memories, further refinement of the retrieval algorithms, and increased adaptability for use across multiple IDEs. These future enhancements would elevate the feature's utility, supporting developers more effectively as Traycer AI continues to evolve its offering.

